Rsearch

A user-friendly R interface to VSEARCH with integrated visualization and automated parameter tuning

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MOTIVATION

VSEARCH[1] is a fast, versatile open-source tool for metabarcoding analysis and a popular choice for high-diversity environmental DNA (eDNA) datasets, where it has been shown to outperform tools such as DADA2 and Deblur[2]. However, its command-line interface may limit accessibility for many researchers. To address this, we developed `Rsearch`, an R package for VSEARCH with integrated tools for visualization and automated parameter tuning.

Rsearch OVERVIEW

`Rsearch` supports core metabarcoding workflow steps, from preprocessing to clustering and taxonomic classification, and extends VSEARCH with visualization, parameter tuning, and data conversion tools.

KEY FEATURES

- Streamlined workflow in R
- Automated parameter tuning
- Integrated visualization
- Modular use of selected components
- Flexible data handling
- Easy integration with other R packages

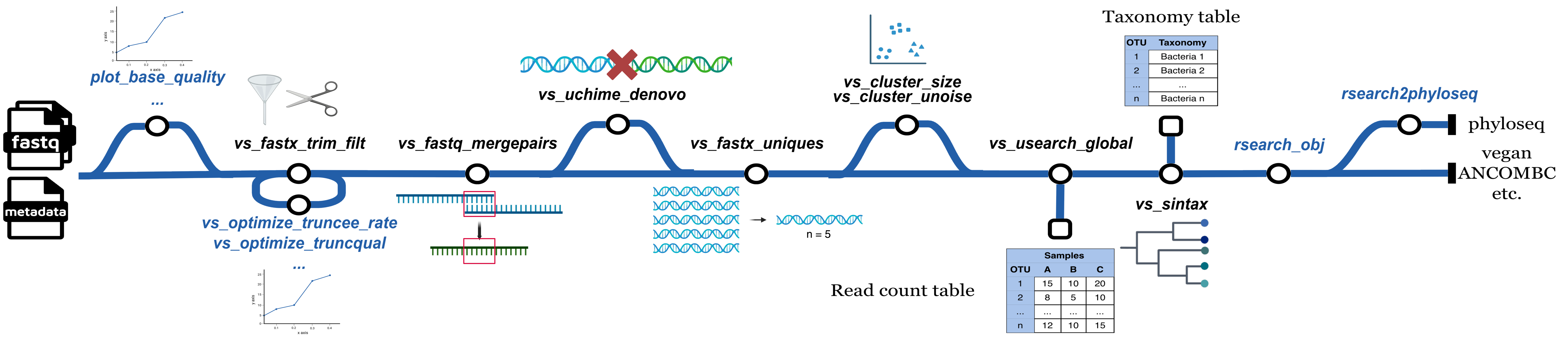

Figure 1. *Example workflow for paired-end sequence analysis with `Rsearch`. The main path represents a typical metabarcoding pipeline, while detours indicate optional steps for visualization or automated parameter tuning.*

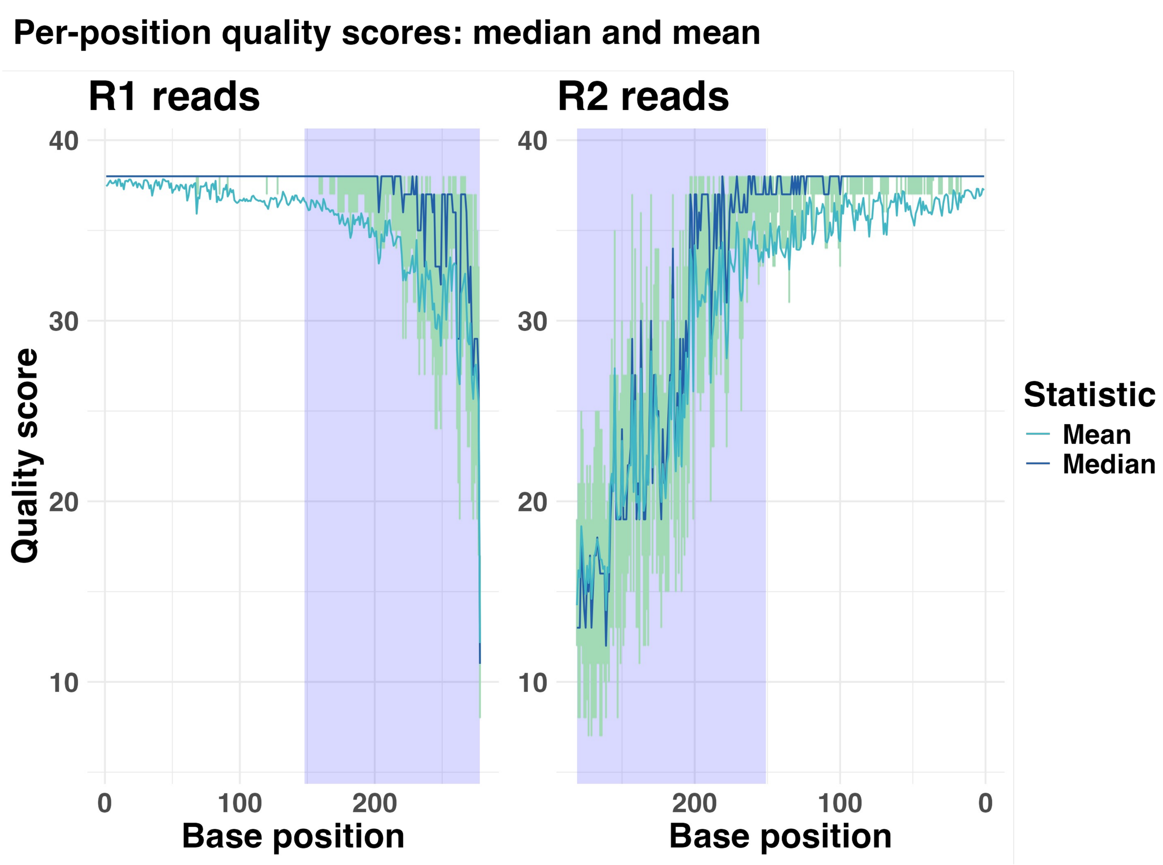

Figure 2. *Inspect paired-end read quality with plot_base_quality. Shaded regions indicate read-pair overlap and help guide the 3′ end trimming. This is a new feature of typical quality profile plots.*

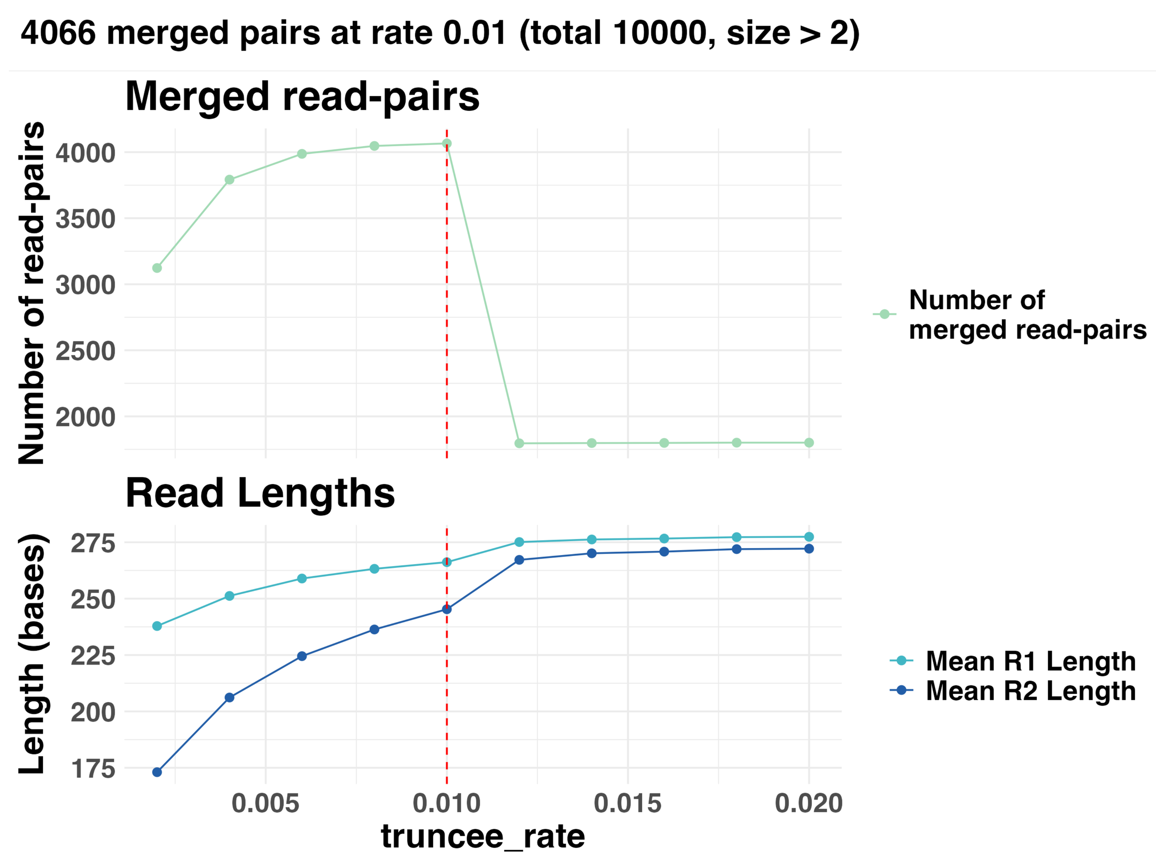

Figure 3. *Automated tuning with vs_optimize_truncee_rate. truncee_rate truncates the 3′ end of reads to keep the average expected error per base below a specified threshold. The red line marks the optimal value: the one that maximizes merged read pairs with copy number ≥2, assuming sequencing errors are more common in singletons.*

BENCHMARKING

Benchmarking against DADA2[3] on a 16S rRNA mock community dataset showed that both pipelines produced accurate abundance profiles but differed in diversity estimates: DADA2 consistently over-split sequences, inflating low-prevalence OTUs, whereas `Rsearch` produced fewer, more prevalent, and more stable OTUs.

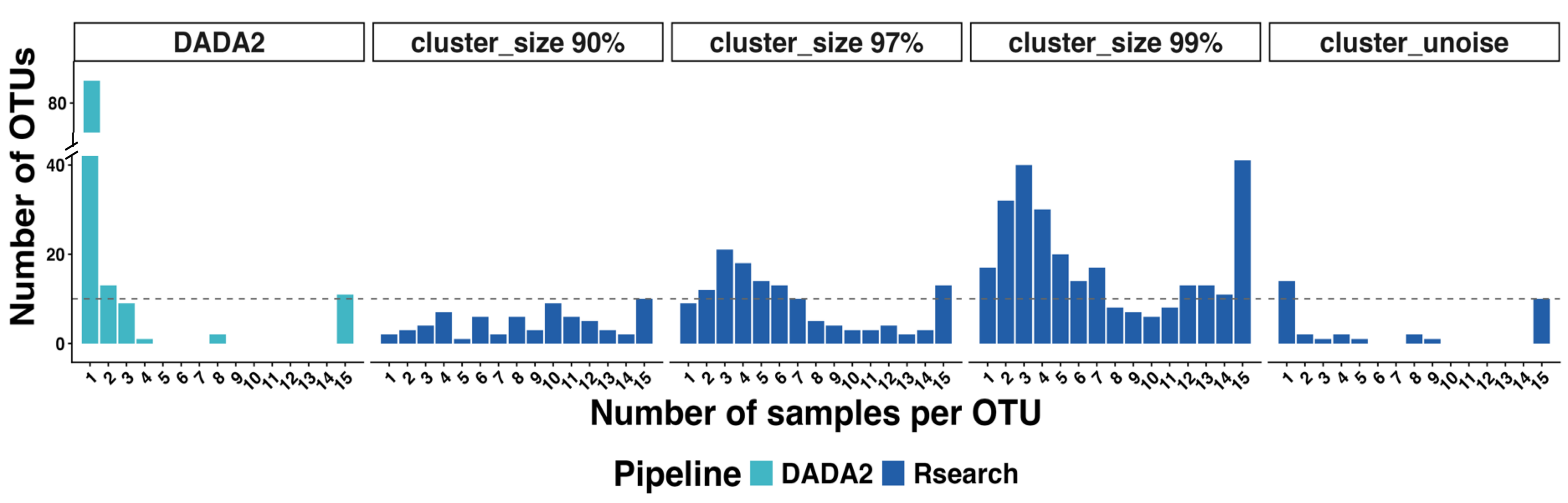

Figure 4. *OTU sample prevalence. The horizontal dashed line indicates the expected 10 OTUs present across all 15 mock community samples.*

CONCLUSIONS

`Rsearch` combines the efficiency of VSEARCH with the accessibility and flexibility of R, providing a reproducible and user-friendly framework for metabarcoding analysis. With integrated visualization and automated parameter tuning, it streamlines key workflow steps and supports analysis of high-diversity datasets.

CRAN GitHub Tutorial

Explore Rsearch here

REFERENCES **1** Rognes, T., Flouri, T., Nichols, B., Quince, C. & Mahé, F. VSEARCH: a versatile open source tool for metagenomics. PeerJ 4, e2584 (2016). **2** Nilsen, T. et al. Swarm and UNOISE outperform DADA2 and Deblur for denoising high-diversity marine seafloor samples. ISME Commun. 4, ycae071 (2024). **3** Callahan, B. J. et al. DADA2: High-resolution sample inference from Illumina amplicon data. Nat Methods 13, 581–583 (2016).

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